

Geometry Basics

AI Tutor - Mathematics - Lesson 3

Introduction

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Explanation

Geometry is the branch of mathematics concerned with shapes, sizes, angles, and the space they occupy. Every shape is built from basic elements: points, lines, and angles. Understanding a few core shapes (triangles, squares, circles) and how to measure perimeter (the distance around a shape) and area (the space inside it) unlocks the ability to reason about the physical world, from room layouts to land surveying.

Key Points

- A triangle has 3 sides and its angles sum to 180 degrees.
- A square has 4 equal sides and 4 right angles (90 degrees each).
- A circle's points are all equidistant from its center.
- Perimeter is the distance around a shape; area is the space inside it.

Examples

- Rectangle perimeter = $2 \times (\text{length} + \text{width})$
- Rectangle area = $\text{length} \times \text{width}$
- Circle area = $\pi \times \text{radius squared}$

Summary

Geometry studies shapes, angles, perimeter, and area, using triangles, squares, and circles as building blocks for more complex spatial reasoning.

Practice Questions

- 1 Find the perimeter of a rectangle with length 8 and width 5.
- 2 Find the area of a square with side length 6.
- 3 Is a 45-degree angle acute, right, or obtuse?

Quiz

Q1. How many degrees do a triangle's angles sum to?
a) 90 | b) 180 | c) 270 | d) 360

Q2. What is the formula for a rectangle's area?

a) length + width | b) $2 \times (\text{length} + \text{width})$ | c) length x width | d) length / width

Q3. An angle greater than 90 but less than 180 degrees is called:

a) Acute | b) Right | c) Obtuse | d) Straight